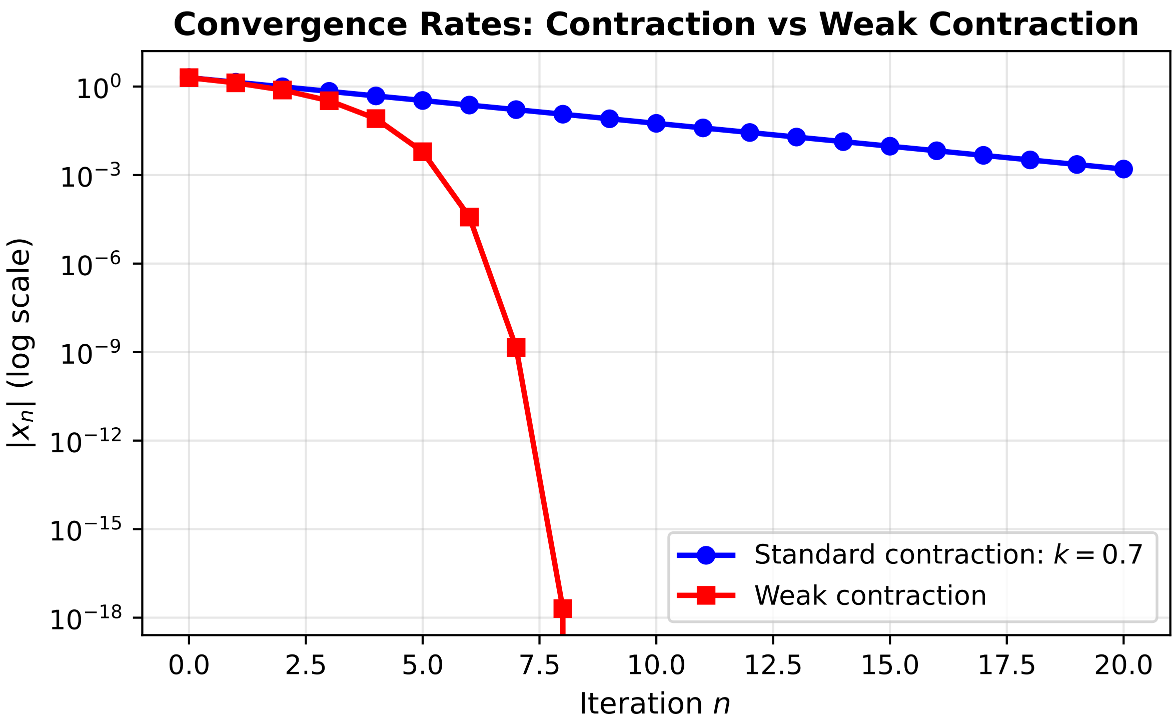

Weakly Contractive Mapping

Definition: For every x, y in X,

||Tx - Ty|| <= ||x - y|| - psi(||x - y||)

where psi: [0, infinity) -> [0, infinity) satisfies:

- psi(0) = 0
- psi continuous and nondecreasing
- psi(t) > 0 for all t > 0
- lim(t->inf) psi(t) = infinity

Examples: psi(t) = at, psi(t) = ln(1+t)